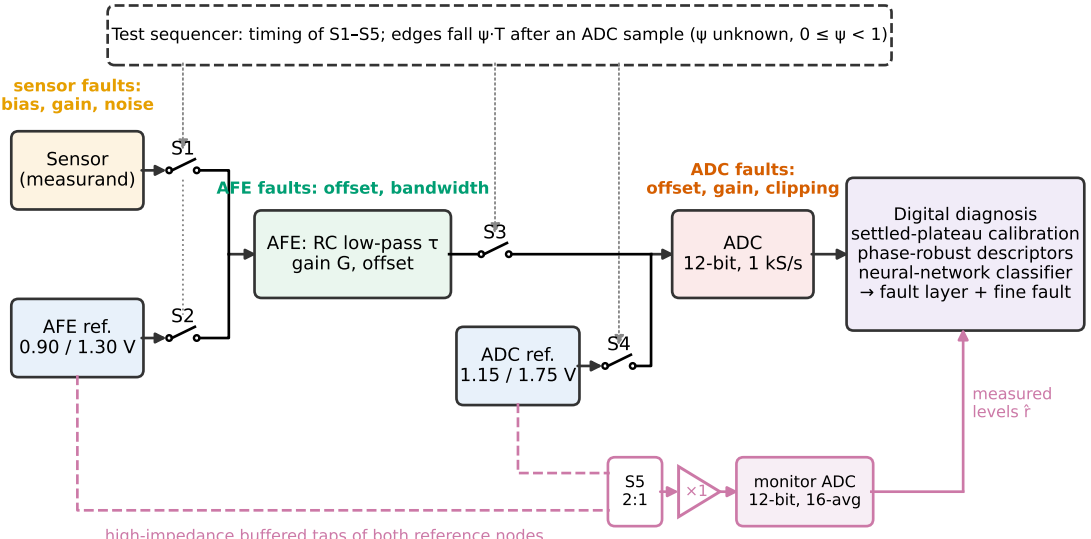
Test sequencer: timing of S1–S5; edges fall ψ·T after an ADC sample (ψ unknown, 0 ≤ ψ < 1)
sensor faults:
bias, gain, noise
Sensor
(measurand)
S1
AFE faults: offset, bandwidth
AFE: RC low-pass τ
gain G, offset
S3
ADC faults:
offset, gain, clipping
ADC
12-bit, 1 kS/s
Digital diagnosis
settled-plateau calibration
phase-robust descriptors
neural-network classifier
→ fault layer + fine fault
AFE ref.
0.90 / 1.30 V
S2
ADC ref.
1.15 / 1.75 V
S4
measured
levels r̂
S5
2:1
×1
monitor ADC
12-bit, 16-avg
high-impedance buffered taps of both reference nodes